

(Task 1 of 10) In this tutorial, we will learn even more about definitions.

0

(Task 2 of 10) What is the result of running this program?

Python | Python [Run 

Pseudo

```
def f(x):          fun f(x):
    return 3       return 3
print(f(7 / 0))    end
                  print(f(7 / 0))
```

⚠ JavaScript behaves differently.

error

2

You predicted the output correctly 🎉🎉🎉

3

Function calls bind their formal parameters (in this case, there is one formal parameter, `x`) to the values of actual parameters (in this case, there is one actual parameter, `7 / 0`). The program errors when it tries to evaluate `7 / 0`.

Click [here](#) to run this program in the Stacker.

(Task 3 of 10) What is the result of running this program?

Python | Python [Run 

Pseudo

```
x = y + 1    let x = y + 1
y = 2        let y = 2
print(x)     print(x)
print(y)     print(y)
```

error

5

You predicted the output correctly 🎉🎉🎉

6

The first definition tries to bind `x` to the value of `y + 1`. To evaluate `y + 1`, we need the value of `y`. But `y` is not bound to a value at that moment.

Click [here](#) to run this program in the Stacker.

Python | 

(Task 4 of 10) What is the result of running this program?

Python [Run 

Lispy

```
x = 12 / 0 (defvar x (/ 12 0))  
print(3) 3
```

 JavaScript behaves differently.

3

8

The answer is error.

9

Textual explanation

You might think `12 / 0` is not evaluated because `x` is not needed. However, this is not the case. In SMoL, every variable definition evaluates the expression immediately and binds the variable to the value, even if the variable is not used later in the program.

Click [here](#) to run this program in the Stacker.

Python | 

What is the result of running this program?

Python [Run 

Scala 3

```
b = 79 / 0 val b = 79 / 0  
print(1) println(1)
```

 JavaScript behaves differently.

error

11

You predicted the output correctly 

12

(Task 5 of 10) In what order are definitions and expressions evaluated?

13

top to bottom

14

(Task 6 of 10) - Variables are bound to values. Specifically, every variable definition

15

evaluates the expression immediately and binds the variable to the value, even if the variable is not used later in the program; every function call evaluates the actual parameters immediately and binds the values to formal parameters, even if the formal parameter is not used in the function.

- Every block evaluates its definitions and expressions in reading order (i.e., top-to-bottom and left-to-right).

Any feedback regarding these statements? Feel free to skip this question.

16

(You skipped the question.)

17

(Task 7 of 10) Please scroll back and select 1-3 programs that together make these points. 18

- Variables are bound to values. Specifically, every variable definition evaluates the expression immediately and binds the variable to the value, even if the variable is not used later in the program; every function call evaluates the actual parameters immediately and binds the values to formal parameters, even if the formal parameter is not used in the function.

You don't need to select all such programs.

(You selected 3 programs)

19

Okay. How do these programs ([4](#), [7](#), [10](#)) support the point?

20

they bound the variables to the values

21

(Task 8 of 10) Please scroll back and select 1-3 programs that together make these points. 22

- Every block evaluates in top-to-bottom, left-to-right order.

You don't need to select all such programs.

(You selected 3 programs)

23

Okay. How do these programs ([1](#), [4](#), [7](#)) support the point?

24

They evaluate top to bottom and left to right order

25

(Task 9 of 10) Here is a program that confused many students

Python | Python [Run]

JavaScript

<code>def addy(x):</code>	<code>function addy(x) {</code>
<code> return x + y</code>	<code> return x + y;</code>
<code>s = addy(1)</code>	<code>}</code>
<code>y = 2</code>	<code>let s = addy(1);</code>
<code>print(s)</code>	<code>let y = 2;</code>
	<code>console.log(s);</code>

Please

1. Run this program in the stacker by clicking one of the [\[Run !\[\]\(9bfa69b6b0f097b09744337d04f22d78_img.jpg\)»» Next](#) until you reach a configuration that you find particularly helpful;
4. Click [Share This Configuration](#) to get a link to your configuration;
5. Submit your link below;

https://smol-tutor.xyz/stacker/?syntax=Python&randomSeed=smol-
tutor&hole=%E2%80%A2&nNext=2&program=%0A%28def fun+ %28add+ x%29%0A++ %28%29

27

(Task 10 of 10) Please write a couple of sentences to explain how your configuration explains the result(s) of the program.

28

The program errors because a variable is called before it was assigned

29

Let's review what we have learned in this tutorial.

30

- Variables are bound to values. Specifically, every variable definition evaluates the expression immediately and binds the variable to the value, even if the variable is not used later in the program; every function call evaluates the actual parameters immediately and binds the values to formal parameters, even if the formal parameter is not used in the function.

- Every block evaluates its definitions and expressions in reading order (i.e., top-to-bottom and left-to-right).

You have finished this tutorial 🎉🎉🎉

Please print the finished tutorial to a PDF file so you can review the content in the future. **Your instructor (if any) might require you to submit the PDF.**

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